GUIDEWIRE DEVTrails 2026 · SEED PHASE

GigShield

Parametric Income Protection

for India's Q-Commerce Delivery Riders

Automatic. Instant. Zero Claims.

PERSONA

Q-Commerce

Blinkit · Zepto

Team CodeNex (SRMIST):
Dhruv Gupta
Aditya Khandelwal
Subhrodeep Ghosh
Astha Chakraborty
Parth Nath Chauhan

THE PROBLEM

India's 3.5 Lakh Q-Commerce Riders Have Zero Financial Safety Net

₹60,000

Lost annually per Delhi rider

20–30% of total income

3.5 Lakh

Active Q-Commerce riders

Blinkit + Zepto combined

₹25,000cr

Market size at risk

Q-Commerce TAM

0

Existing products

No insurer covers this today

Why traditional insurance fails this segment:

- No payslip, no Form 16 — gig workers can't prove income
- 30–90 day settlements are useless when rent is due tomorrow
- No product exists that covers environmental income loss for gig workers
- Monthly premiums don't match week-to-week gig cash flows

Why Q-Commerce Riders Face Extreme Vulnerability

Attribute	Q-Commerce (Blinkit/Zepto)	Food Delivery (Zomato/Swiggy)
Delivery radius	1.5–3 km from dark store	5–10 km from restaurant
Trips per day	20–40 trips	8–15 trips
Time per trip	8–12 minutes	25–45 minutes
Vehicle	Bicycle / e-bike	Motorcycle
Disruption impact	**BINARY — zero income instantly**	Partial — can slow down

3×
more outdoor exposure hours

10min
SLA = binary income cutoff

Zone Lock
Can't migrate to safer area

Bicycle
Can't ride through floods

Weekly
Income cycle needs weekly cover

THE SOLUTION

How Parametric Insurance Works

TRADITIONAL

Event › Claim Filed › Documentation › Investigation › Assessment › Settlement (30–90 days)

GIGASHIELD

Event › Threshold Breached › Auto-Verify external API › Fraud Check AI › **Instant Payout < 30 min**

Claim filing	Manual, with docs	**Zero — fully automatic**
Payout speed	30–90 days	**< 30 minutes**
Income proof	Payslip, employer cert	**None required**
Fraud detection	Post-claim adjuster	**Real-time AI, pre-payout**

WHO WE PROTECT

Real Workers. Real Income at Risk.

Arjun

26 · Rohini, Delhi NCR

Blinkit · Bicycle

Daily earn	Lost
₹1,100/day	₹2,800

⚡ AQI 450+ for 11 days

Without GigShield:

Platform cut assignments 60%. Daily earnings collapsed to ₹280. Could not pay rent.

With GigShield:

AQI > 300 trigger auto-fires. ₹350 payout before end of day. ₹2,800 received over 11 days.

Priya

23 · Andheri West, Mumbai

Zepto · E-bike

Daily earn	Lost
₹1,100/day	₹1,100

⚡ 89mm rain in 6 hours

Without GigShield:

Zone flooded. Operations suspended. Full day's income lost with no recourse.

With GigShield:

Rain > 65mm trigger fires within 45 minutes. ₹450 credited to UPI before rain stopped.

Ravi

29 · Kondapur, Hyderabad

Blinkit · Motorcycle

Daily earn	Lost
₹1,100/day	₹3,300

⚡ 3-day heatwave 46°C+

Without GigShield:

Continued working, suffered heat exhaustion. Lost 3 days recovering. No income protection.

With GigShield:

Heat > 44°C trigger pays ₹350/day. Financial cushion to rest rather than risk health.

PARAMETRIC TRIGGERS

5 Auto-Triggers

Heavy Rainfall

Tier 1

> 35mm / 24hr

Tier 3

> 100mm or IMD Red

₹200 – ₹600

OpenWeatherMap

Mumbai, Kolkata, Pune

Hazardous AQI

Tier 1

> 300 AQI, 4hr+

Tier 3

> 500 or GRAP IV

₹150 – ₹500

CPCB Portal (Govt)

Delhi NCR, Lucknow

Extreme Heat

Tier 1

> 43°C, 3hr+

Tier 3

> 47°C or IMD Alert

₹150 – ₹450

OpenWeatherMap + IMD

Delhi, Ahmedabad, Jaipur

Cyclone / Storm

Tier 1

Wind > 55km/h

Tier 3

> 110km/h + Red Alert

₹250 – ₹750

OpenWeatherMap + IMD

Mumbai, Chennai, Kolkata

Curfew / Bandh

Tier 1

Partial zone order

Tier 3

> 24hr or state bandh

₹200 – ₹700

Mocked (NLP Phase 3)

Pan-India coverage

WEEKLY PREMIUM MODEL

AI-Driven. Affordable. Actuarially Sound.

Weekly Premium = Base Rate × Zone Risk × Seasonality × Coverage Tier × Disruption History

Base Rate	Zone Risk	Seasonality	Coverage Tier
₹20–₹30	1.0–2.8×	1.0–2.0×	1.0/1.4/1.8×
Minimum weekly floor	*Dark store zone history*	*Sin/cos month encoding*	*Basic/Standard/Premium*

City Risk Multipliers

Delhi NCR	2.6×	AQI crisis + heat + civic disruptions
Mumbai	2.4×	Extreme monsoon + cyclone exposure
Kolkata	2.1×	Cyclone proximity + urban flooding
Hyderabad	1.9×	Severe heatwaves + flash floods
Pune	1.7×	Heavy monsoon rain
Bengaluru	1.4×	Moderate — lowest risk

Example calculations

Zepto cyclist · Rohini, Delhi · Standard · Monsoon peak
≈ ₹170/week

Blinkit e-bike · Andheri, Mumbai · Basic · Winter
≈ ₹60/week

Zepto cyclist · Koramangala · Standard · Normal
≈ ₹51/week

Target loss ratio: 65–70% · Phase 2: XGBoost + LightGBM ensemble replaces rule-based formula

AI / ML ARCHITECTURE

Three AI Models.

Model 1 — Dynamic Premium Pricing

XGBoost + LightGBM (60% / 40% ensemble)

- Zone GeoHash + flood/AQI risk tier
- Sin/cos month — monsoon & smog seasonality
- Vehicle type (bicycle vs e-bike vs motorcycle)
- Historical disruption count (12-month rolling)
- Self-reported daily trip volume

Model 2 — Fraud Detection Engine

Isolation Forest + GradientBoosting (scikit-learn)

- GPS satellite variance (spoofed = zero variance)
- Accelerometer RMS — motion vs stationary
- IP geolocation vs GPS zone delta
- Claim velocity vs zone peer median
- Pre-event zone residency window check

Model 3 — Disruption Prediction (Phase 3)

LSTM (PyTorch CPU) · 7-day zone forecast

- Output: P(disruption_event) per zone per week ⟶ feeds reserve planning

SHAP Explainability — Admin Dashboard

Arjun's ₹153/week: Base ₹25 + AQI zone +₹45 + Monsoon +₹38 + Disruption +₹25 + Tier +₹20

TRAINING	DATA	SERVING	TRACKING
Google Colab free T4 GPU	**50K synthetic rows (Faker)**	**FastAPI /predict · .pkl**	**MLflow (Docker, free)**

ADVERSARIAL DEFENSE

Anti-Spoofing Strategy — 4 Independent Signal Layers

⚠️ Syndicate threat: 500 riders coordinating via Telegram, using mock GPS apps to fake locations in disruption zones, draining the liquidity pool.

GPS Physics

- Zero satellite variance ⟶ physically impossible
- Accuracy radius = 0m ⟶ spoofed
- Instant GPS lock ⟶ no satellite search
- Path vs OSM road geometry

Device Sensors

- Accel RMS: cycling 0.8–2.4 m/s²
- Stationary home rider: < 0.3 m/s²
- Gyro yaw vs GPS heading delta
- Android ALLOW_MOCK_LOC flag check

Network Geo

- IP vs GPS delta > 4km ⟶ flag
- Cell tower handoff count = 0 ⟶ flag
- Carrier subnet cross-check
- ip-api.com (1K free calls/day)

Behavioral

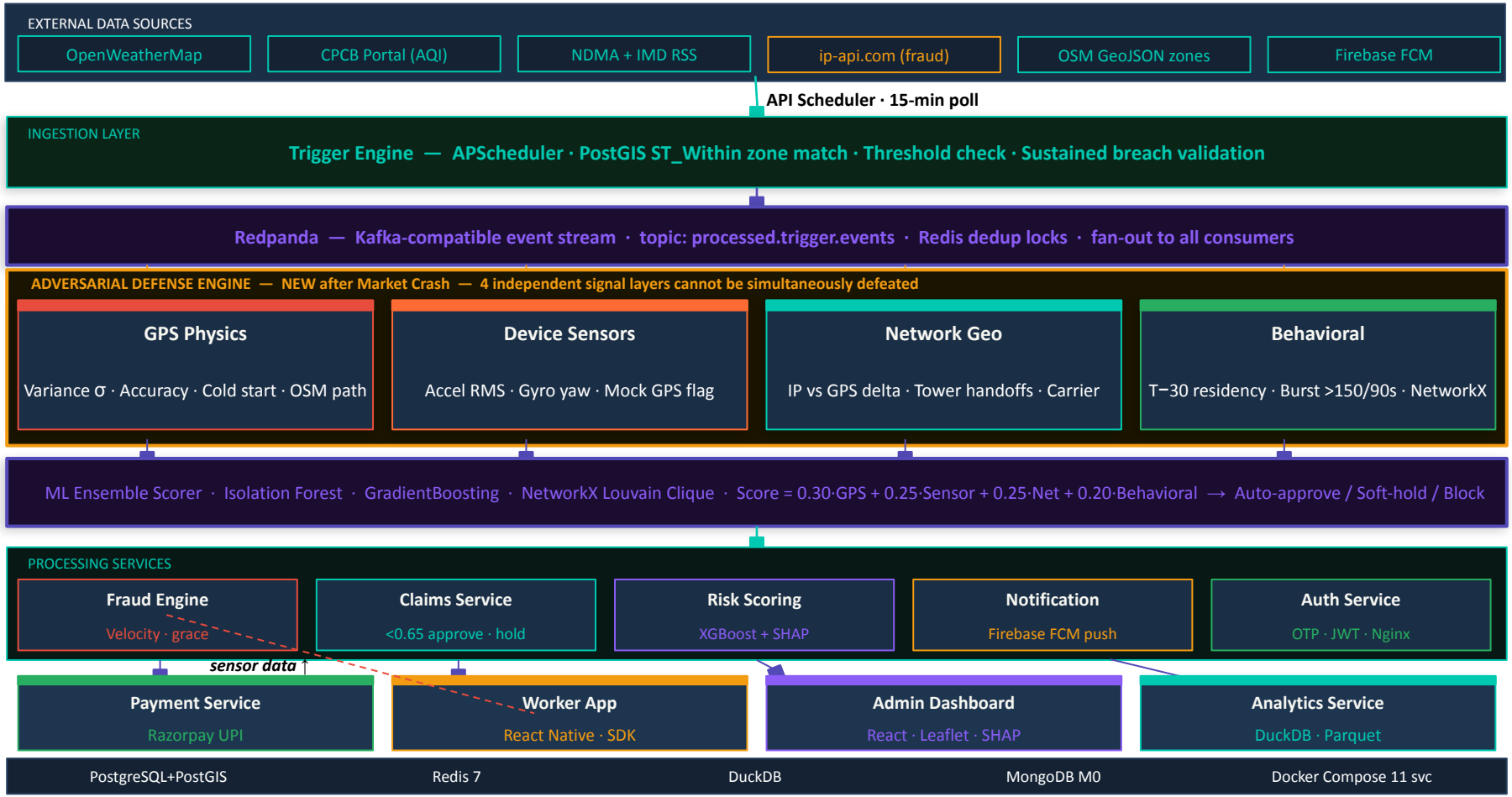
- Pre-event zone residency T−30min
- Burst > 150 claims in 90 seconds
- NetworkX Louvain graph clique detection
- Velocity > 2.5× zone peer median

ML Ensemble Scorer: Isolation Forest · Behavioral GradientBoosting · NetworkX Louvain Clique Density

Score < 0.30	Score 0.30–0.65	Score > 0.65
Auto-approve	**Soft hold**	**Block**
Full payout in < 5 minutes	50% now · 50% held 2 hours	Admin review · 4hr SLA

Network drop grace: 45-min GPS gap forgiveness if last ping was inside the zone — never penalises genuine Zepto cyclists on low-end phones

SYSTEM ARCHITECTURE



TECH STACK

Infrastructure for All 6 Weeks

Cloud: Railway.app free ($5 credit · covers 6 weeks) · Frontend: Vercel free · CI/CD: GitHub Actions free · Training: Google Colab free T4 GPU

Frontend

React Native + Expo Go	Worker mobile app
React + Vite + Tailwind	Admin dashboard
Recharts	Charts & analytics
Leaflet.js + OSM tiles	Risk heatmap

Backend & Data

Python FastAPI	All 11 microservices
Redpanda Docker	Kafka-compatible queue
Supabase free (500MB)	PostgreSQL managed
PostGIS extension	Zone matching (free)

AI / ML

XGBoost + LightGBM	Premium ensemble
scikit-learn Isolation Forest	Fraud ML
PyTorch LSTM (Phase 3)	Disruption forecast
SHAP + MLflow	Explainability + tracking

APIs & Infra

OpenWeatherMap free	1,000 calls/day
CPCB govt portal	Unlimited AQI data
Razorpay test mode	Free UPI simulation
Firebase Auth + FCM	Free OTP + push

TEAM + ROADMAP

5 Builders. 3 Phases. One Mission.

Dhruv Gupta
Team Lead ·
AI/ML · XGBoost + IsoForest + LSTM

Aditya Khandelwal
Backend · FastAPI + Razorpay + Redpanda

Subhrodeep Ghosh
Frontend · React Native + Dashboard

Astha Chakraborty
UX/UI & Architecture Research

Parth Nath Chauhan
Data · APIs + PostGIS + DuckDB

Development Roadmap

SEED — ACTIVE
Mar 4–20

- README.md & persona research
- Docker skeleton + PostGIS zones
- Premium calculator (rule-based)
- 2-min video submission

SCALE — NEXT
Mar 21–Apr 4

- Worker + Policy services
- Trigger Engine (OpenWeatherMap poll)
- Razorpay test mode end-to-end
- React Native worker dashboard

SOAR — PLANNED
Apr 5–17

- XGBoost premium model trained
- Isolation Forest fraud deployed
- Admin dashboard + heatmap live
- 5-min demo video + Pitch Deck PDF

"India's quick-commerce economy promises groceries at your door in 10 minutes. Behind every delivery is a rider with zero financial protection when the sky turns against them."

Zero Claims

Event fires automatically
No worker action needed

Real-Time AI

Fraud blocked pre-payout
Not 45 days later

Weekly Model

Matches gig cash flows
₹35–₹200 per week

No claims. No paperwork. No delays.

Protection that works as fast as the deliveries we depend on.